Generating Embodied Storytelling and Interactive Experience of China Intangible Cultural Heritage “Hua’er” in Virtual Reality

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ABSTRACT

Hua’er, a type of traditional oral performance, is one of the national intangible cultural heritages (ICH) in China. Experts have been trying to enhance public’s awareness of Hua’er protection through digital documentation technology, but there’s no efficacious means to attract interest and popularize knowledge yet. In this paper, we propose an interactive VR system engaging audiences to experience and understand the connotation of Hua’er performance. Based on an online survey with the public and in-depth interviews with Hua’er experts, we derive a set of design requirements for generating embodied storytelling and interactive experience in VR. Accordingly, we design “Hua’er and the Youth” (HY) that integrates three methods (virtual avatar, participatory performance, and game-based knowledge acquisition) and conduct a between-subjects user study to compare HY with the Comparison system. The results suggest that our methods significantly improved audience’s interactive experience, knowledge level and the awareness of ICH safeguarding.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality; • **Applied computing** → Arts and humanities; Performing arts.

KEYWORDS

Virtual reality, intangible cultural heritage, embodied storytelling, game-based knowledge acquisition, Hua’er

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1 INTRODUCTION

According to United Nations Educational, Scientific, and Cultural Organization (UNESCO), the intangible cultural heritage (ICH) refers to various traditional cultures that exist in intangible form, are closely related to the lives of the people and inherited from generation to generation. ICH [1, 2] includes oral traditions and expressions, performance art, social practices (rituals and festive events), traditional craftsmanship, etc. Among them, traditional oral performance is an important part of ICH, such as Maloya (France), Flamenco (Spain), Gagaku (Japan), and Hua’er (China) [3]. Due to its nonphysical nature, ICH is more restricted and harder to demonstrate in real life which is a challenge to protection work. The researchers have been working on the digital preservation and inheritance through advanced technology for years [4-6]. However, how to increase people’s attention to ICH and achieve its widespread dissemination is still an urgent problem.

In Gansu, Qinghai Provinces and throughout north-central China, nine minorities share a musical performance known as Hua’er [7, 8]. Hua’er is a type of folk song performance, which usually be sung in dialects in the gathered “Hua’er Festival”. Because of historical, natural, and human factors, Hua’er is in decline. Specifically, first, the original ecological living environment was destroyed and gradually disappeared. Second, the number of existing Hua’er inheritors is gradually decreasing. Third, the general public, especially the young generation, have little knowledge of the historical value and humanistic value of ethnic regional music culture. The ecological and cultural environment in which Hua’er survives is becoming fragile [9].

With the rapid development of digital technology, the Hua’er digital resource library and related website appear, and become the main way for safeguarding and dissemination [11-14]. However, for general public, this passive watching experience may not be effective. It is thus of necessity to explore new dissemination paradigm to present Hua’er and educate the general public in interactive way [15, 16]. There has been a rising trend of virtual reality (VR) in the ICH field, with the advantage in scenario construction, non-linear narrative and active interaction [17-25]. We would like to incorporate digital materials into VR technologies to generate embodied storytelling and interactive experience for ICH dissemination. This is currently underexplored.

In this paper, we designed "Hua'er and the Youth" (HY), an interactive VR prototyping system to involve audience in a virtual storytelling experience of one of the most popular Hua'er song "Baxiguliuliu". We integrated methods that involving embodied cognition and game-based learning to turn the audience from passive recipients into active participants. Through a between-subjects user study with 33 participants, we compared HY either with or without these methods. The results suggest that HY remarkably improve audiences' sense of VR and interactive experience, as well as consciousness of Hua'er protection and dissemination. Besides, the methods are also effective in terms of transmitting knowledge. The primary contributions of our work are as follows:

(1) a VR design approach integrating virtual avatar, participatory performance, and game-based knowledge acquisition to overcome the challenges in generating embodied storytelling and active learning of Hua'er in a virtual environment.

(2) design, technical and methodological implications on generating Hua'er knowledge and experience employing VR derived from user studies. Our methodology and insights are not unique to Hua'er only but can be employed by any other oral performance of ICH.

2 RELATED WORK

2.1 Hua'er Conservation in China

Recognized as a world intangible cultural heritage since 2009, the Chinese government has put significant effort into protecting the intangible cultural relics via establishing Hua'er research academy and conservation institutions [8, 10]. For the conservation of Hua'er, which includes physical and digital methods. Physical conservation concerns teaching and inheriting the specific skill of Hua'er that generally spreads through word of mouth. Digitization now is one of the most important methods for ICH dissemination. Currently, nearly 50 Hua'er performances and "Hua'er Festival", a large-scale folk singing competition which is held at a specific time and place, have gradually begun to be recorded and disseminated to the public [9]. The Hua'er digital resource library has been online since Jun 2021. However, current dissemination methods, including the documentary, the film, exhibition, even VR-based application, still have some weaknesses, for instance, it is difficult for the public to understand the culture behind Hua'er based on the immobilized performance content. The public also reported that current approaches were insufficient in terms of experience and engagement.

2.2 Cultural Heritage in VR

VR technologies enable 3D visualization and 3D interaction modalities, allowing the public to get immersive experience [29, 30]. Currently, research on ICH in VR is mainly focused on digital reproduction of the museum-like, visualized ICH scenes [22, 26, 31–33]. Luo [34] realized the visual simulation of Sichuan opera performance, which the audience can interact with the virtual environment through gestures, bodies, etc. With the development of HCI in VR, researchers have begun to make use of the digital narrative to create new methods for disseminating ICH, while emphasizing the interaction with the audience [32, 35, 36]. For example, Kim et al [19] proposed concept for designing content of Korea Traditional Dance ExperienZone (IKTDEZ) based on target user and interactive storytelling. These approaches are also embodied activities [37], which emphasized the role of audience participation [27, 28, 38]. Currently, introducing virtual avatars in a VR environment is one of the ways to implement embodied storytelling in the ICH experience, which can be widely used in ICH digital museums, historical and cultural venues, or even serious games [39-41]. Meanwhile, VR can also enhance user's participation and entertainment through game-based learning experiences. Dagnino et al [42] developed the "*Canto a Tenore game*", supporting the basics of this rare singing style learning to involve the user in an engaging journey. The gamified experiences for ICH in VR usually pay more attention to non-material content, and provide a high degree of audience participation. However, the effectiveness of serious games for ICH dissemination requires in-depth research.

3 "HUA'ER AND THE YOUTH": CONCEPT AND IMPLEMENTATION

Our design goal was to explore effective design methods, allowing the audience to participate in the immersive experience, enhancing their understandings of the connotation of Hua'er. Therefore, we proposed "Hua'er and the Youth", an interactive VR system for intangible heritage experience. We firstly distributed 226 online questionnaires including 20 questions among the general public to investigate their interests and sources of information about the oral performance in ICH, in specific, their understanding level of Hua'er, the attitude towards the combination of VR and ICH, the willingness to experience more about ICH, and what requirements and characteristics the VR experience should have. Results showed that the public understanding level of Hua'er is limited, and the combination of ICH and VR is applicable. Furthermore, the public also shows a serious lack of knowledge of Hua'er and its historic culture. The existing Hua'er performances seem difficult to attract young people's interest. Thus we refined the visual and singing formal characteristics of Hua'er and increased the interactive experience to encourage audience to be more involved in the performance. We constructed a new paradigm of virtual narrative that combining oral performance of ICH with virtual reality.

3.1 Design Concept and Material

According to the requirements from pre-investigation, we found that virtual characters and participatory interaction are important in VR narrative, and the gamified interaction could bring more entertainment [41, 43, 44]. The virtual avatar should accompany with visual effects, actions, dialects to better present the local performance. As for the participatory interaction, it should allow the audience to join the oral performance and become part of it. Moreover, the system should provide gamified experience to enhance the dissemination of the historical and cultural background of "Hua'er" in an interesting way. Therefore, in "Hua'er and the Youth", we designed and implemented "*virtual avatar*", "*participatory performance*", and "*game-based knowledge acquisition*". We chose one of the classic songs of Hua'er, "Baxiguliuliu", and took the participation in the "Hua'er Festival" as the main story line. The system includes three scenes, namely "An Invitation", "The Festival" and

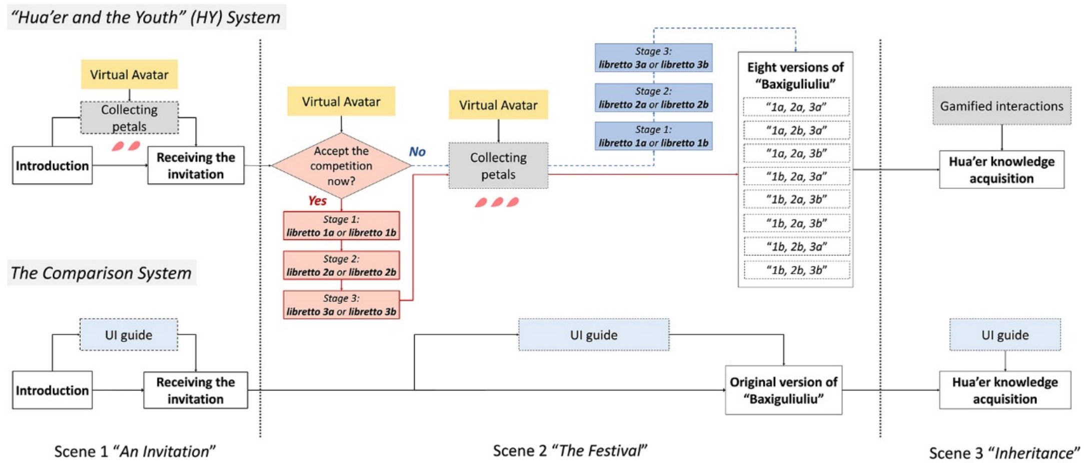

Figure 1: The design structure of HY

“Inheritance” (see Figure 1). We collected the digital materials of “Baxiguliuliu” included 11 performance videos and 10 original audios that recorded by Hua’er inheritors. Due to the characteristics of multi-tone and multi-word in one song of Hua’er, we synthesize eight different versions of “Baxiguliuliu” as adaptive feedback according to the audience’s choice of narrative. Meanwhile, we integrated the historical background knowledge and folk culture knowledge of Hua’er that audience are interested in into the scenes, as an illustration, we collected digitized Hui people’s costumes and paper-cutting materials of "Northern Faction" in Ningxia. Moreover, we added virtual avatar to guide the audience to complete the task of participatory performance and learn Hua’er knowledge. The design structure of HY is shown in Figure 1.

3.2 VR Scenario and Interaction Design

We designed three virtual scenes in “Hua’er and the Youth”, scene 1 “An Invitation”, scene 2 “The Festival”, scene 3 “Inheritance” to allow the audience experience “Baxiguliuliu” and learn Hua’er knowledge. When entering scene 1, the audience were guided to identify their roles and were introduced what Hua’er is. Then, the audience received an invitation letter and prepared to go to the “Hua’er Festival”. In scene 2, the audience can trigger a dialogue with the male avatar and learn about the meaning of "Baxiguliuliu" according to his commentary. Then, the audience would receive virtual petals as knowledge cues, and trigger different performance on the stage in different storylines. In scene 3, audience needs to complete the fill-in-the-blank game on a book to learn Hua’er knowledge. In Comparison system, the virtual environment and performance content are basically the same as in HY. The difference is that we remove the virtual avatars, participatory performance, and gamified interactions. The audience is guided by conventional UI to arrive at “Hua’er Festival” and watch the original version of “Baxiguliuliu” under the stage. For knowledge acquisition, the audience can gain the information only by reading a book.

3.2.1 Virtual Avatar. In HY, we designed two virtual avatars, the audience avatar and a companion avatar. ***(1) Audience avatar:*** we draw lessons from the visual characteristics of Ningxia Hui people’s paper-cutting craft to design a female character to represent the audience’s role, who is a young Hua’er inheritors. We designed the hand models in paper-cutting form to enhance the audience’s sense of identity with the virtual character. We extracted three patterns from female costumes in Hui, namely, large flowers at the jacket, cuff, and trousers, then we refined these visual features (Figure 2a) and applied them to the female avatar (Figure 2b). ***(2) Companion avatar:*** the male character has a double identity-an experienced Hua’er inheritor and a guide. The male avatar could explain Hua’er knowledge to the audience by guiding them to collect virtual petals, and help audience complete the duet performance. We extracted the patterns on both sides of the waistcoat chest, placket and trousers (Figure 2c), forming the paper-cutting body of him (Figure 2d).

3.2.2 Participatory Performance. The audience can decide which time point to participant in the performance (before or after collecting petals) by accepting the competition invitation (Figure 2e) from the companion avatar. If the audience accepts, the companion avatar will guide the audience to randomly choose librettos three times with different rhymes to accomplish a complete “Baxiguliuliu” track which can evolve into 8 different versions (Figure 2f). When arriving at the “Hua’er Festival”, the stage setting will present the complete track just composed by the audience. If the audience does not accept firstly, the companion avatar will guide the audience to collecting petals, then invite the audience to choose preferred librettos to achieve a complete track (Figure 2g). As for dialect words and metaphor tactics of Hua’er, we set up symmetrical librettos between dialect and mandarin (such as “好尕尕” and “*Good Singer*”) to help understand. We also reappeared the virtual objects mentioned in the librettos, such as “*hanging on the chest of the pouch*”, “*flowers sprout when they fall to the ground*”, etc, to convey the meaning of the librettos visually.

3.2.3 Game-based Knowledge Acquisition. Game-based knowledge acquisition refers to knowledge learning by means of game-based

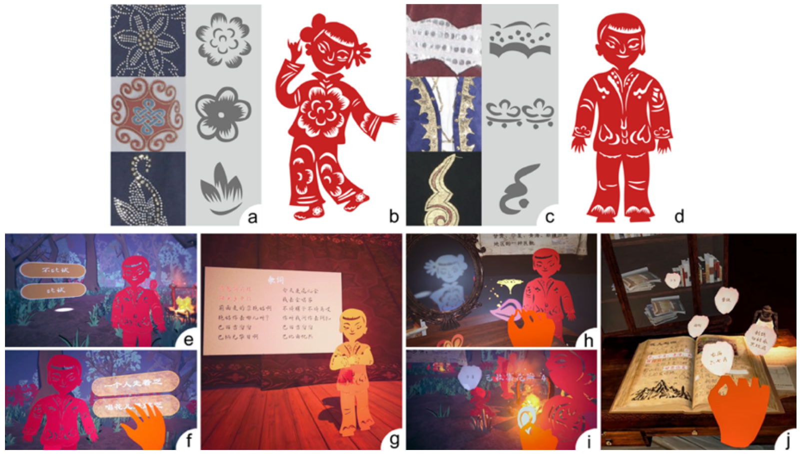

Figure 2: (a) Patterns from female costumes in Hui. (b) The female avatar. (c) Patterns from male costumes in Hui. (d) The male avatar. (e) Accepting the competition. (f) Choosing the librettos. (g) Singing with the companion avatar in a duet. (h) Collecting the first petal. (i) Collecting the fourth petal. (j) The fill-in-the-blank game for Hua'er knowledge acquisition .

interaction [46]. In HY system, the audience can acquire the popularization knowledge of Hua'er through dialogues with the companion avatar, petals collection and the fill-in-the-blank game. We integrated the knowledge into the narrative of "Baxiguliuliu", specifically in two steps: **1)** In scene 1 and 2, the audience collected five knowledge cues presented as virtual petals (Figure 2h. 2i). **2)** In scene 3, the audience needs to put five petals in the corresponding positions on a book to acquire all Hua'er knowledges (Figure 2j). If all correct, a scaled-down female avatar with songs will appear as a reward. In the *Comparison* system, the audience will passively acquire Hua'er knowledge by reading scrolls UI or automatically turning pages of the book.

4 USER STUDY

We conducted a between-subjects user study with 33 participants in *comparison* HY with the *Comparison* system. In this study, the independent variables (IV) were HY and its *Comparison* system. The dependent variables (DV) were selected to examine our design goal of generating embodied storytelling and interactive experience of oral performance "Baxiguliuliu" from Hua'er through interactive VR. Our DVs were split into four aspects, i.e. 1) VR experiencing; 2) interactive experience; 3) Hua'er knowledge acquisition; 4) overall awareness of ICH dissemination and safeguarding. At the end of the user study, we also conducted in-depth interviews with the participants. We hypothesized:

H1. Compared to the *Comparison* system, HY provides significantly more immersive (***H1a***) and enjoyment (***H1b***) VR experience.

H2. HY provides better interactive experience with significantly more embodied cognition (***H2a***), entertainment (***H2b***), fluent (***H2c***) interaction than the *Comparison* system.

H3. Game-based knowledge acquisition generated by HY is significantly more effective for enhancing audiences' knowledge level (***H3a***) and awareness on ICH dissemination (***H3b***) than the *Comparison* system.

The study was carried out in a university laboratory. The participants were 18 males and 15 females with the average age of 24.10 (*SD* = 3.90). They were randomly divided into two groups, HY and *Comparison* system, namely 17 participants (P1-P17) for HY and the other 16 for the *Comparison* system (P18-P33). Participants of HY or *Comparison* put on the headset and earphones to experience the system. In these two experiments, the whole knowledge cues were presented in same order and the content was basically uniform. The variable was that the participants of HY experienced the performance with embodied storytelling and gamified interaction, while the *Comparison* did not. We carried out the study via watching participants on the computer monitor and recording encountered problems during the experiment. After the experience, the participants had to complete four questionnaires, and took a semi-structured interview. All studies on HY for one participant lasted about 25 minutes, and those of *Comparison* lasted around 18 minutes.

5 RESULTS

We first analyzed the distribution of all DVs' scores with the Shapiro-Wilk test. Apart from enjoyment, interaction entertainment, and individual knowledge acquisition, all other DVs were distributed normally. The Welch's T-test was used to analyze all DVs' scores that distributed normally, and the Mann-Whitney's U test was used for the DVs not distributed normally, both of which analyzed non-paired samples. Figure 3 showed the overview of the final experimental results.

5.1 VR Experience

The E^2I evaluation method had been widely used to evaluate VE's immersion, enjoyment, and engagement by subjects [45]. The E^2I total scores and subscale scores for presence were analyzed by the Welch's T-test. Participants in HY reported a significant higher enjoyment score (M = 5.90, SD = 0.98) compared with *Comparison* (M = 5.27, SD = 0.95), (Z = -2.178, p = 0.029).

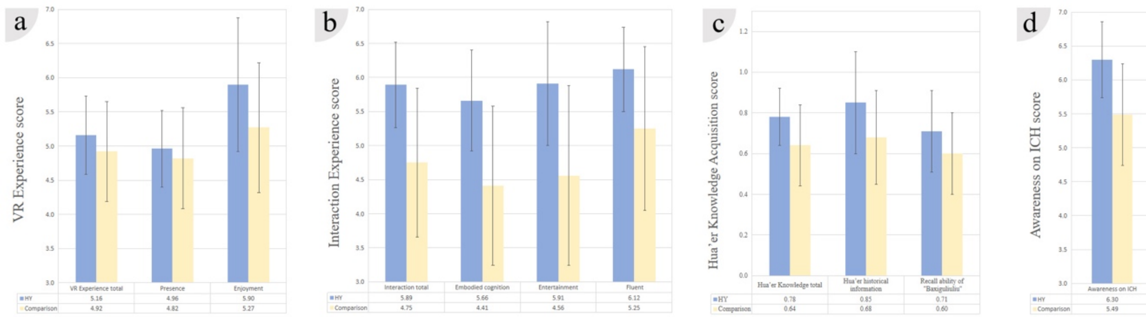

Figure 3: Results of user study with standard deviation. (a) E^2I, presence and enjoyment. (b) Interaction experience with embodied cognition, entertainment and fluent. (c) Hua’er knowledge acquisition. (d) Awareness on ICH dissemination.

5.2 Interactive Experience

We evaluated the embodied cognition, entertainment and interaction fluency for each system by using the 7-point Likert scale questionnaire. For embodied cognition, the analysis was conducted by the Welch’s T-test, and the result showed that HY’s total interaction experience on (M = 5.89, SD = 0.63) was significantly more higher than that of *Comparison* (M = 4.75, SD = 1.09), (t(31) = 3.692, p = 0.001). Participants in HY also reported significantly higher entertaining interaction experience (M = 5.91, SD = 0.91) in *comparison* with that of *Comparison* (M = 4.56, SD = 1.32), under an analysis with the Mann-Whitney’s U test (Z = -3.283, p = 0.001).

5.3 Hua’er Knowledge Acquisition

We set seventeen questions that involved basic Hua’er historical information(Q1-Q8) and the recall questions of “Baxiguliuliu”(Q9-Q17) to test the participants’ knowledge level after the system was put into use. The results showed that participants in HY reported a significantly higher total knowledge score (M = 0.78, SD = 0.14) than Comparison’s participants (M = 0.64, SD = 0.20), (t(15) = 2.279, p = 0.038). However, the statistic results indicated that the participant’s knowledge level of Hua’er historical information and recall ability of “Baxiguliuliu” was not significantly difference between HY and Comparison system. Specifically, HY (M = 0.85, SD = 0.25), (M = 0.71, SD = 0.20), Comparison (M = 0.68, SD = 0.23), (M = 0.60, SD = 0.20), (t(15) = 1.934, p = 0.072), (t(15) = 1.681, p = 0.114). Further, we found the scores of Q5, Q8, and Q12 in HY (M = 0.64, SD = 0.49), (M = 0.65, SD = 0.49), (M = 0.47, SD = 0.51) were significantly higher than those in Comparison system (M = 0.25, SD = 0.45), (M = 0.19, SD = 0.40), (M = 0.06, SD = 0.25), under an analysis with the Mann-Whitney’s U test (Z = -2.254, p = 0.024), (Z = -2.629, p = 0.009), (Z = -2.591, p = 0.01).

5.4 Awareness on ICH Dissemination

To assess the participants’ awareness for ICH dissemination, we set six 7-point Likert scale questions, including the evaluation of awareness on Hua’er performance, willingness for further dissemination, and attitude to interactive VR approach to experience and learn ICH. The total score was analyzed by a Welch’s T-test, and participants of HY reported a significantly higher score (M = 6.30, SD = 0.56) compared to *Comparison* (M = 5.49, SD = 0.75) (t(31) = 3.538, p = 0.001).

6 DISCUSSION

Our research objective was to evaluate the impact of VR experiencing, Hua’er knowledge acquisition, interactive experience, and overall awareness on ICH dissemination in HY. The overall E^2I and presence scores indicated that HY was not provided significantly more immersive experience than *Comparison* (***H1a*** rejected). However, it produced significantly more enjoyment (***H1b*** accepted) than *Comparison*. This result could be interpreted as, first, we did not make major visual changes to the *Comparison* system. Second, although we added various interactions to the HY system, we found our current system was not doing enough for user feedback.

In regarding to the interactions and procedures, we found significantly difference in the rating of total interactive experience, embodied cognition (***H2a*** accepted), entertainment (***H2b*** accepted), and fluent (***H2c*** accepted) between the two systems. Most participants in HY reported that they were encouraged a strong sense of identity when they saw their avatar. As (P8) said, “*I think it’s very interesting to interact with paper-cutting people in ICH performances, especially when I see myself as paper-cuts when I sing with the actor on stage, I am still very impressed.*” Moreover, the participants in HY were highly evaluated the interaction fluency. Most participants said that although they did not know how to interact at the beginning, they gradually became proficient through the guidance of the male avatar and the sounds of the surroundings. Therefore, the participants reported that the overall interaction experience is very smooth.

While according to the interviews, our system still had room for improvement. For example, (P10) mentioned that “*although the paper-cutting avatar is traditional and inherited, the immersive experience may not be so strong.*” (P12) said that “*most people have limited access to ICH inheritors in reality and would be interested in performers wearing traditional costumes.*” (P17) from the Hui nationality who showed his interest in the current form of virtual avatar. But he also pointed out that combining digital humans with paper-cutting avatars may be a more suitable way to appreciate Hua’er performance in VR. This showed another potential problem in our system, which was the lack of a mechanism to encourage and reward the participants.

Then, in terms of the participatory performance, we found that a few participants did not realize that the composed song they enjoyed was chosen by themselves. This mainly happened because the participants had to choose the librettos at the very beginning, after collecting the petals, the participants were easily forgot the content that they selected before. We considered two approaches for this weakness: first, we could set up some adaptive feedback to the librettos when the participants chose them; Second, we could design specific interactions that can correspond to the selected librettos during the composed performance. For example, when the selected librettos are played on stage, the virtual crowd reacts with cheers or praise.

As for the Hua'er knowledge level, participants in HY reported a significantly higher scores of the total Hua'er knowledge compared with the *Comparison* system (***H3a*** accepted), which improved that our design methods were more effective for enhancing audiences' knowledge level. We also found that HY got a slightly higher score than *Comparison* of Hua'er historical information and the recall ability of "Baxiguliuliu". However, there was not significantly difference between them. The possible reason was that the methods of virtual avatar and the game-based interaction did not have an absolute influence on increasing the knowledge level and there might be some unreasonable settings when we design the questions. Finally, we found that HY improved ICH dissemination awareness significantly compared to *Comparison* system (***H3b*** accepted).

7 CONCLUSION

In this paper, we presented "Hua'er and the Youth", an interactive VR system that engages audience to experience oral performance of Chinese ICH, Hua'er, in an immersive environment. The HY system was designed through the online survey with the public and in-depth interviews. Our design goal was to generate embodied storytelling and interactive experience in VR to attract interest and popularize knowledge to the general public, especially, the young generation. To examine the design objective and evaluate HY, we conducted a between-subjects user study with 33 participants to compare our system with the Comparison system. The results suggested that HY significantly increased participants' enjoyment and interactive experience, as well as their awareness of ICH dissemination over the proposed methods. Furthermore, the game-based interaction was more effective than ordinary reading in the aspect of transmitting knowledge. In all, we found that multi-avatar interaction, particular folk knowledge learning, and advanced gamified experience might give rise to further attention and willingness to other oral performance dissemination that needs to popularize to the public.

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